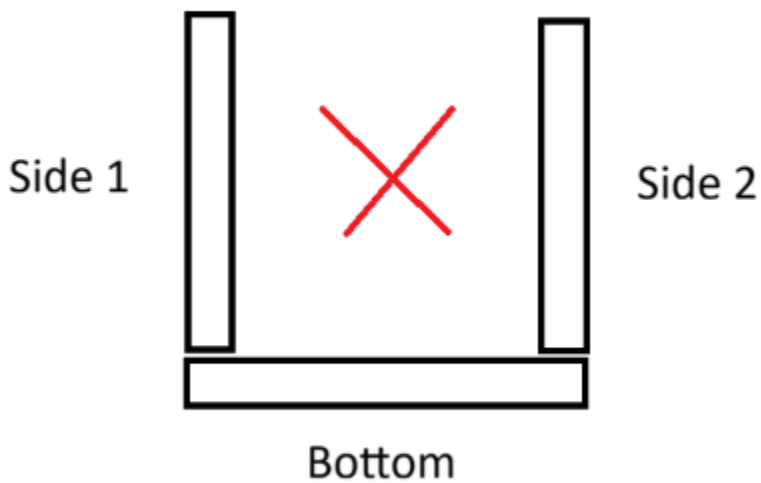
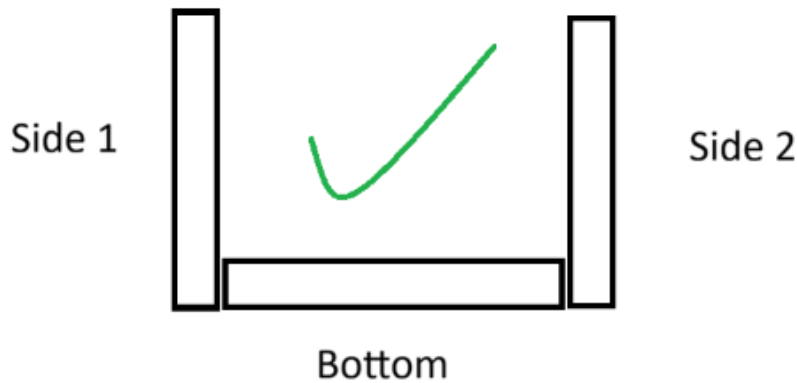


All Dimensions are in centimeters(CM)

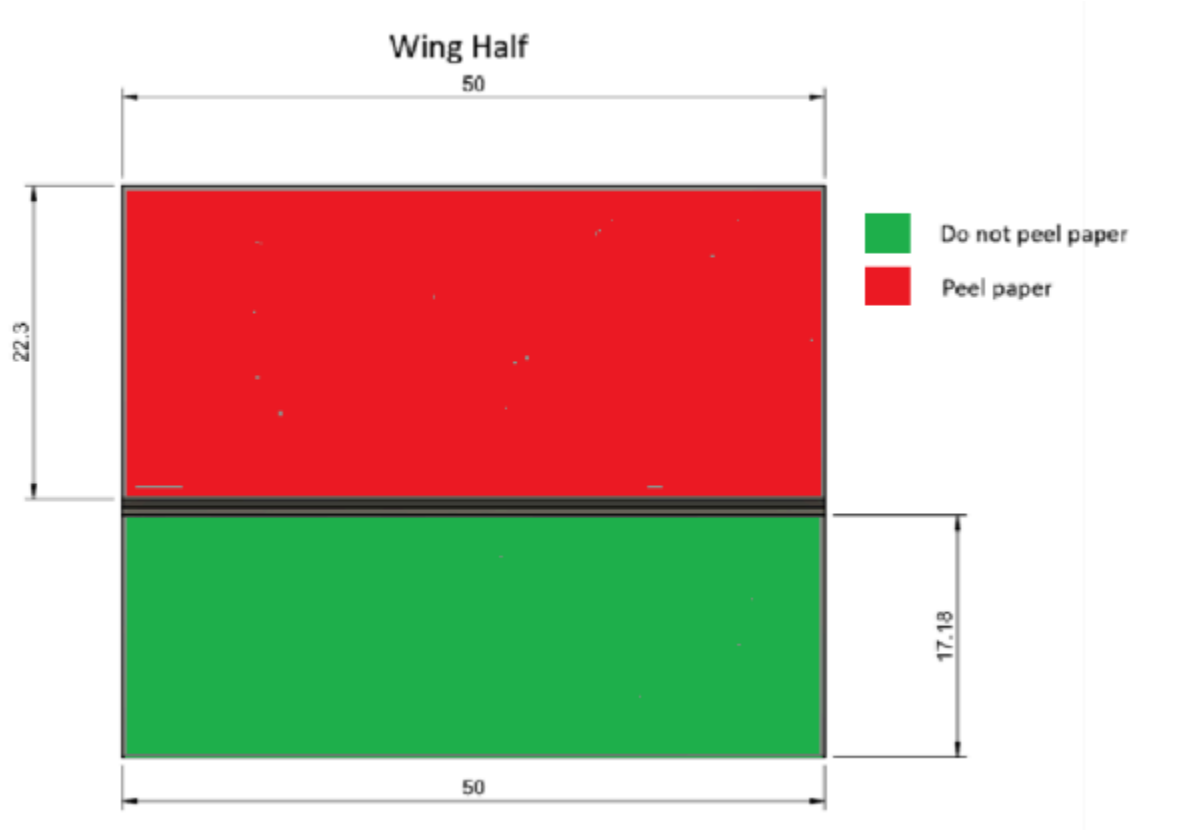
Step 1: Fuselage

Firstly you need to assemble the airframe. Cut out all the parts from the blueprints section of the github repository. It is best to cut out the 2 sides and the bottom piece first. After cutting, glue them so that the two sides sandwich the bottom in between them. Make sure you do not put the sides on top of the bottom piece.

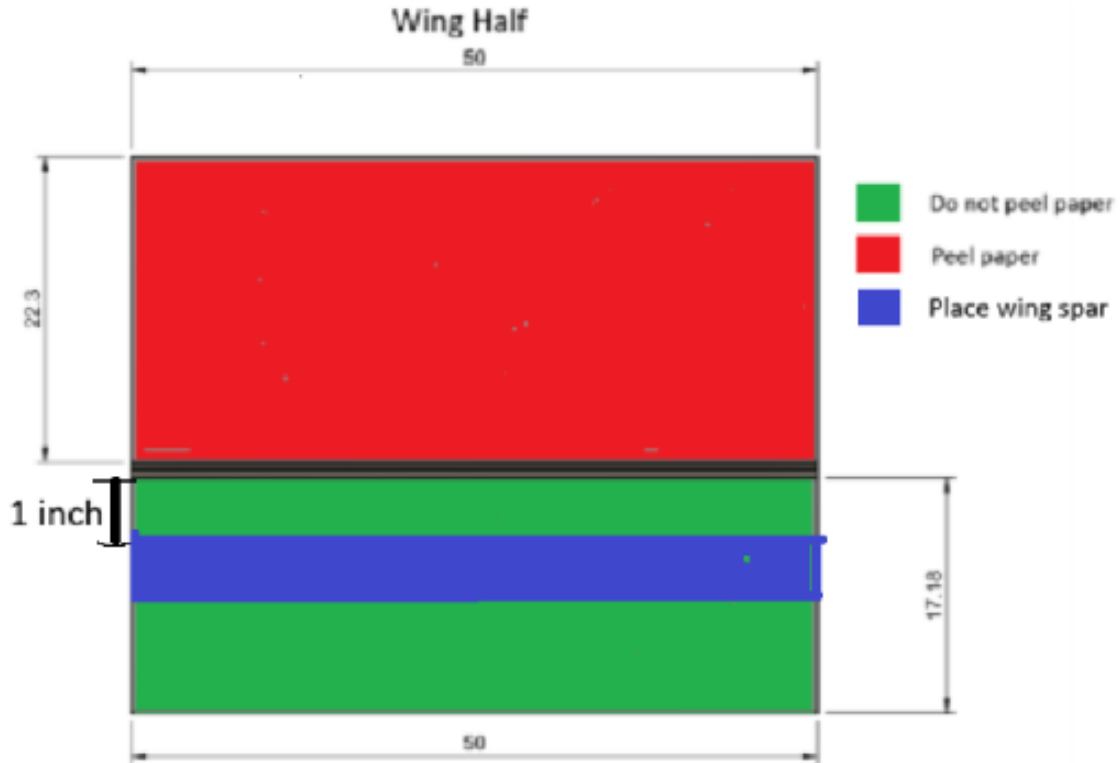


Step 2: Wing

Next cut out the “wing” pieces. Both halves require the same size piece of foam. Also cut out 2 “wing spars” for each wing half and glue both spars together. Once the two spars are glued together, set it aside and peel the paper off the 22.3 x 50 cm area off both wing halves. If the paper is not peeling off properly, try to use 91% isopropyl alcohol to dissolve the glue.



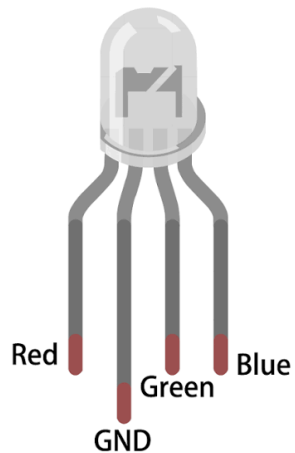
Next place and glue the wing spar onto the highlighted area. It is one inch back from the crease in the non-peeled area.



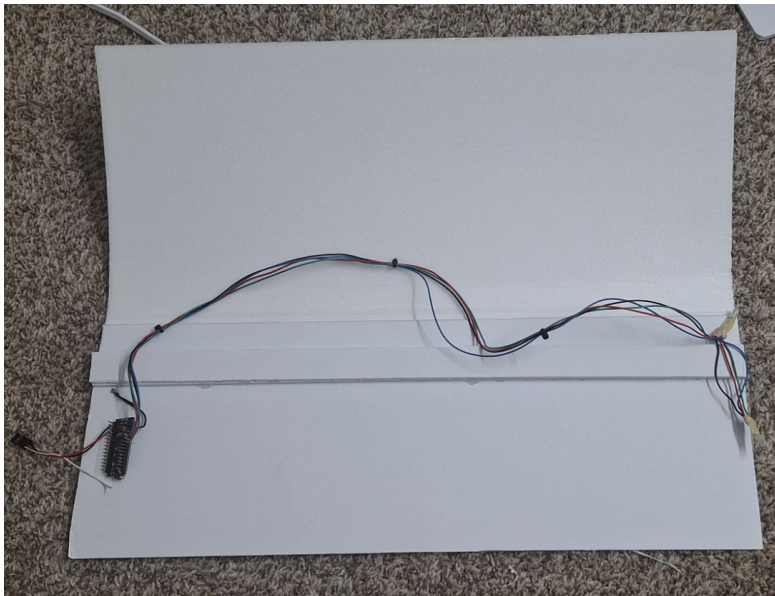
This next part is an optional aesthetic upgrade to add strobe lights to your plane. Note: *It does not significantly affect the plane's stability, responsiveness and aerodynamic efficiency.*

Skip to Page 5 if you wish to not install any lights.

You need 1 programmable board that is compatible with arduino IDE (An arduino nano is recommended for the most lightweight lights system) , 6 5v RGB leds, 6 200 ohm resistors and 4 rolls of differently colored 26 AWG wire(Red, Black, Green and Blue). Run red, blue, green an black wires to the tail of the plane and solder them. Black wire the GND, Red wire to pin 2, Green wire to pin 3, Blue wire to pin 4. Now solder other ends of the wire to the leds and make sure the colors match. Black always goes to the cathode(The longest leg) via a 200 ohm resistor.



Next run a red and black wire through the wing and solder 2 leds to the wire. Black to GND and red to red on led and pin 4 on arduino. Repeat with a black and green wire on the right side.

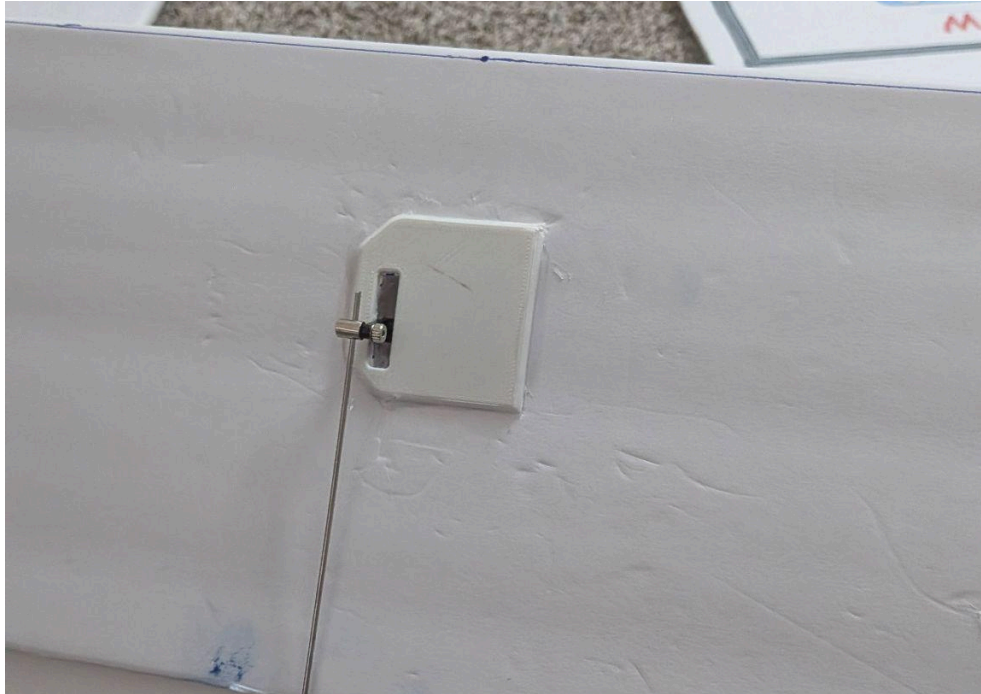


Note: Aircraft navigation lights use a standard color code and placement: a red light on the left wingtip, a green light on the right wingtip, and a white light on the tail

Finally connect a 3 pin connector (like the ones found on your servos) and cut off the signal wire. Connect the red wire to arduino 5v and the black wire the arduino's second GND pin.

Step 2: Wing (Continued)

Now install the servo mount into the bottom of the wing by printing the “servo mount.stl” and using the 2 screws that were included in the packet to install the servo. Next cut a hole in the bottom of the wing and slot in the servo mount and glue it in place. Next attach the pushrod in place.



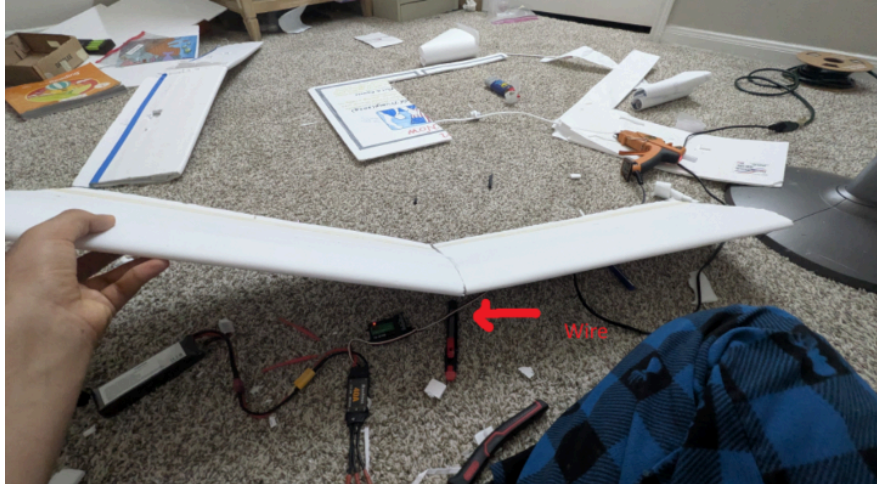
Now crease the line when the peeled and non-peeled section meet and fold the peeled section over the wing spar and non peeled area like in this video:

[Wing Building Demo](#)

Make sure you apply hot glue to the top of the wing spar prior to this.

Repeat with the right side as well but make sure the flat part of both the servo mount faces the fuselage. ***If the servos move together the plane will not roll. They should be placed in opposite orientations so that when they are connected with a y cable to a single channel on the receiver one goes up the other goes down.***

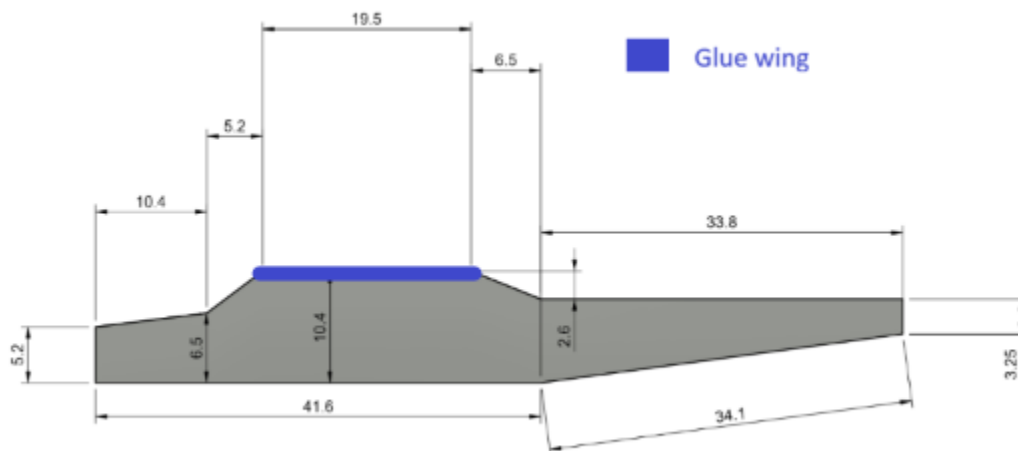
After constructing the other half, route all the wires out of each wing half so that they all come out of the bottom of where the two halves meet. Next 3d print the “15 degree dihedral.stl” as the connecting piece between the wing halves and glue it in place.



Now cut out 2 more wing spars which will be used as ailerons. Cut one of their edges at 45 degrees as shown in the video: [How to cut at 45 degrees](#)
Do this to both ailerons. Then attach them to the trailing edge of each wing using duct tape. Also use the duct tape to reinforce the leading edge and the middle where the 2 halves meet to get the wing completed.

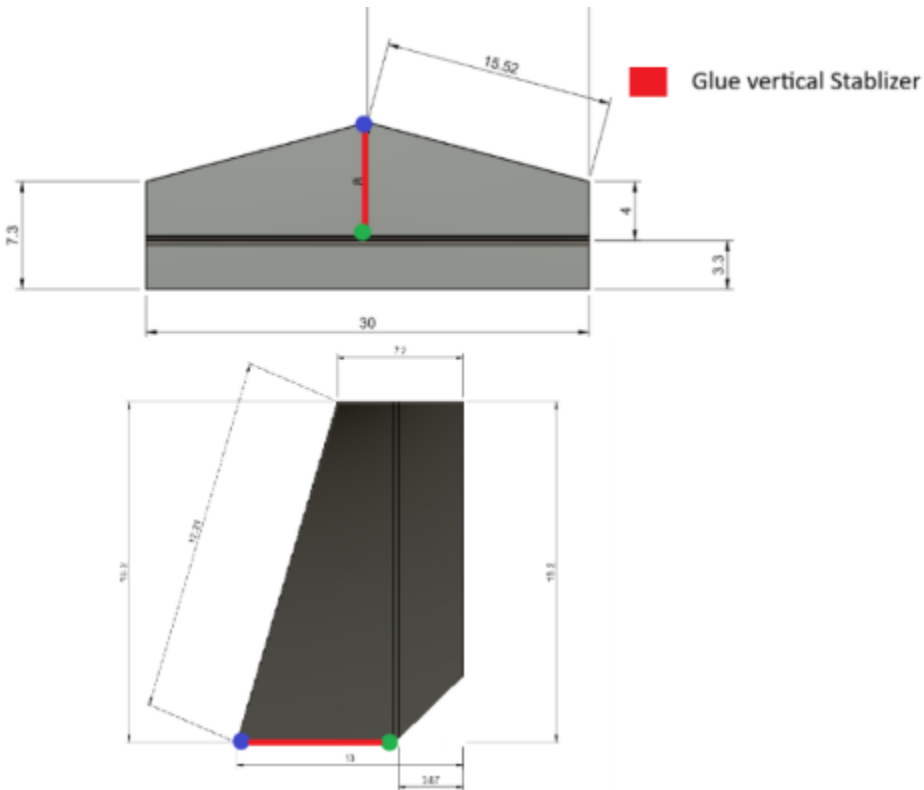


Now glue it to the fuselage making sure it is centered and aligned.

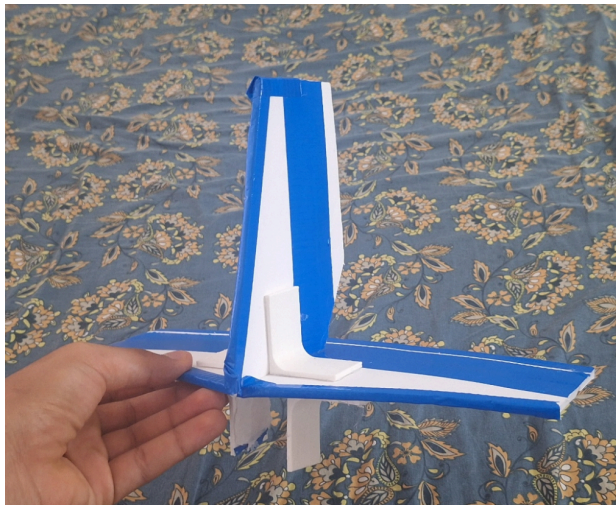


Step 3: Tail

Now cut out the “Horizontal stabilizer” and the “Vertical Stabilizer”. Now glue the vertical stabilizer on to the horizontal stabilizer. Making sure they follow the center line and line up the dots shown:



Put duct tape all over the tail and print out 4 “Rudder and Elevator brace.stl” and glue them onto the tail to reinforce them like such:



Finally glue the tail onto the end of the fuselage and connect your last servo to the elevator/horizontal stabilizer to complete the airframe!



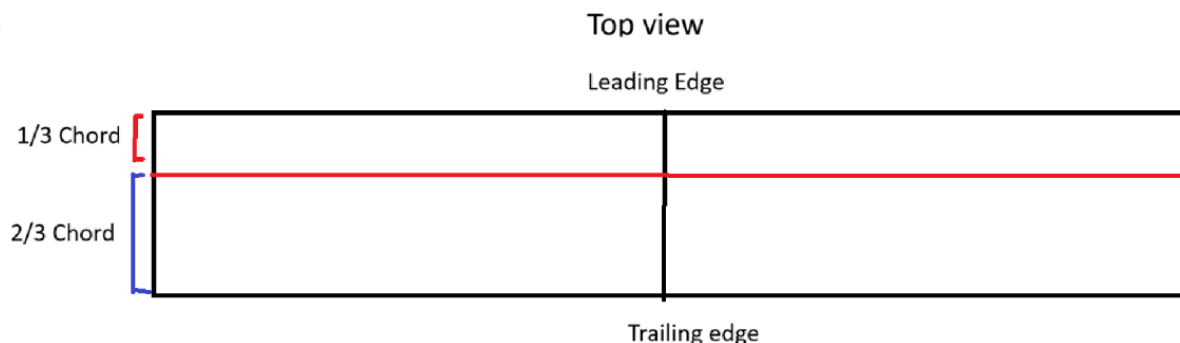
Step 4: Electronics

print the “motor mount.stl” and attach your 2122 motor to it using 3 m3 nuts and screws. Connect all 3 wires into your ESC and make sure it is spinning the correct way. If it is spinning in the wrong direction, swap any 2 wires and it will reverse. Once again, the ESC 3 pin connector goes to channel 3 on the receiver.

The Y harness splitting the signal to the 2 servos in the wing goes into channel 1 in the receiver and the elevator/horizontal stabilizer servo goes into channel 2. Also if you did choose to add lights, connect the 3 pin connector coming from the arduino into any available channel except for channel 1-4. Connect your 2200 mah 3s lipo to the esc and secure it with velcro or tape to make sure it does not slide during flight.

Step 5: Balancing the plane

Balancing the plane is the single most important thing to ensure the plane flies well or even flies at all. Hold the plane on fingers and slide the battery back and forth so that it balances perfectly $\frac{1}{3}$ the way back from the leading edge.



Step 6: Miscellaneous

You are almost ready to fly! Just print out the “front panel.stl” and the “NACA duct to scale.stl” glue them together and secure it to the front of the plane to cover the electronics bay. It is best to use rubber bands to secure it. Put the same duct tape used previously on the bottom of the fuselage to waterproof it and help with belly landings.

Step 7: Pre-Flight Checks

Your plane is now ready to fly! Just perform these checks to make sure your plane flies well.

- Make sure the battery is secure so it does not slide around.
- Tighten the propeller. If it falls off in flight, the plane is doomed :)
- Check all control surfaces and make sure they are moving
- Make sure the battery is charged
- Place the antenna of the receiver outside the plane to ensure good signal.

Flight Tips

- When taking off, set the motor to 100 percent to gain airspeed fast
- Throw it at a shallow angle to avoid a stall
- Get used to the controls in a simulator before you fly the plane.

Flight Videos

- [Landing Video](#)
- [Take Off Video](#)